

UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS



Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

RESEARCH PROTOCOL [INITIAL DRAFT]

DELIVERABLE No. 3

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DELIVERABLE No. 3

Research topic: A Human-in-the-Loop Multimodal AAC Prototype in Spanish to Support Functional Communication in Autistic Children Requiring Very Substantial Support

Protocol note. This deliverable now presents the full ten-component protocol structure reviewed in class. Sections 02 (Abstract) and 04 (Literature Review) are included as preliminary development sections and will be completed and refined in subsequent sessions, particularly after the systematic literature review and PRISMA-based synthesis.

01. Title

A Human-in-the-Loop Multimodal AAC Prototype in Spanish to Support Independent Functional Communication in Autistic Children Requiring Very Substantial Support

02. Abstract

Student-authored abstract scaffold. In the final version, this section should briefly synthesize the communication problem, the specific research gap, the objective of the human-in-the-loop multimodal AAC prototype, the Design Science Research approach, the bounded evaluation context and primary outcomes, and the expected scientific contribution. The abstract will be completed after the Literature Review and the subsequent refinement of the protocol so that it accurately reflects the final study design and evidence base.

- Background / problem: functional communication barriers among minimally speaking or nonspeaking autistic children requiring very substantial support.
- Research gap: limited evidence on culturally and linguistically localized, human-in-the-loop, multimodal AAC personalization in Spanish-speaking settings.
- Objective and method: design and evaluate one bounded AAC prototype under a DSR build-evaluate-iterate logic.
- Evaluation focus: independent functional communication, communicative diversity, and assistance/prompting required.
- Expected contribution: evaluated artifact plus transferable design principles for responsible, context-aware personalization.

03. Introduction & Problem Statement

Many autistic children who are minimally speaking or nonspeaking and require very substantial support experience persistent barriers to expressing needs, preferences, discomfort, choices, emotions, and requests for help through speech alone. Augmentative and alternative communication (AAC) can provide an alternative channel, but the practical usefulness of an AAC system depends on whether its vocabulary, interaction flow, symbols, and output remain understandable, relevant, and controllable by the user.

In the Peruvian context, the challenge is compounded by the limited availability of Spanish-language AAC solutions that can be adapted to family routines, local vocabulary, and sociocultural context. Static AAC interfaces can be useful, but they may require caregivers or professionals to anticipate in advance which messages will be relevant in a specific interaction. A multimodal, context-aware system could potentially reduce this limitation by suggesting communication options based on available contextual information; however, such personalization also introduces risks related to privacy, bias, over-automation, homogenization of communication, and loss of user agency.

The scientific problem is therefore not simply whether artificial intelligence can be added to AAC, but how a human-in-the-loop multimodal AAC artifact can be designed so that contextual personalization improves functional communication without speaking on behalf of the child or reducing the child's control. The current evidence base does not, by itself, establish how such an artifact should be designed and evaluated for a clearly bounded Spanish-speaking user group in a structured therapeutic context. This gap motivates a Design Science Research (DSR) study centered on building, evaluating, and refining a functional prototype under explicitly limited conditions.

To keep the doctoral project feasible within three years, the study will focus on one clearly defined subgroup of minimally speaking or nonspeaking autistic children requiring very substantial support, one structured therapeutic communication context, and two to three primary communication outcomes. The study will seek rigorous pilot evidence and transferable design knowledge rather than broad claims of clinical effectiveness.

04. Literature Review

Preliminary evidence map. Two recent 2026 studies are incorporated at this stage because they directly inform the technological and evaluation space of the proposed thesis. The narrative literature review will be expanded in subsequent sessions through a systematic search, screening, and synthesis process.

Table 1

Recent Evidence on AI-Enabled and Context-Aware AAC Relevant to the Proposed Thesis

Study	Design / sample	Main finding	Gap / relevance to this thesis
Griffen et al. (2026)	Three autistic preschool children; ABAB reversal design; context-aware AAC prototype compared with traditional AAC for academic communication.	The context-aware condition produced more frequent, accurate, and faster communication responses across participants.	Shows that automated, context-aware AAC can improve performance, but the study is small, focused on academic communication, and calls for broader and longer-term evaluation. It does not establish Spanish/Peruvian localization, multimodal human-in-the-loop personalization, or the broader functional-communication outcomes proposed here.
Pitt et al. (2026)	Preliminary AI/AAC study using DeepFace; 12 neurotypical children and 4 autistic children aged 8-13; individualized calibration for emotion recognition.	Promising provisional accuracy and generally positive user experience, with substantial variation by emotion and participant.	Supports personalization and AI-enabled AAC, while explicitly identifying needs for multimodal inputs, active AAC users, more diverse developmental/cultural samples, and spontaneous real-world interactions. These gaps align with the proposed thesis direction.

Note. Author’s own elaboration based on Griffen et al. (2026) and Pitt et al. (2026). The table synthesizes study design, sample, main findings, and research gaps relevant to the proposed thesis.

- The recent evidence strengthens the rationale for AI-enabled AAC, but it also narrows the novelty claim: the thesis should not claim novelty merely because it uses AI, context awareness, multimodality, autism, or a single-case comparison.
- The more defensible contribution is the integrated design problem: a Spanish-language, culturally contextualized, human-in-the-loop multimodal AAC artifact focused on independent functional communication, communicative diversity, and assistance requirements while explicitly preserving user agency and safety.
- The DSR contribution should be framed as both an evaluated artifact and transferable design principles for responsible contextual personalization, rather than only as an accuracy or performance improvement.
- This preliminary literature section will be extended after the systematic review / PRISMA activity in the following sessions.

05. Research Questions, Hypotheses & Objectives

Main research question: How can a human-in-the-loop multimodal AAC prototype in Spanish be designed and rigorously evaluated to support independent functional communication in one clearly defined group of minimally speaking or nonspeaking autistic children requiring very substantial support, within one structured therapeutic context?

Table 2

Research Questions of the Proposed Study

Code	Research question
RQ1: Design	How can the prototype provide contextually relevant communication options while preserving user agency, personalization, and the right to reject or ignore suggestions?
RQ2: Evaluation	To what extent does the prototype improve independent functional communication, communicative diversity, and the level of assistance required when compared with a predefined static AAC condition?

RQ3: Risk & ethics What privacy, autonomy, bias, homogenization, and safety risks emerge during use, and what safeguards are required to manage them responsibly?

Note. Author's own elaboration. The research questions are organized into three dimensions: artifact design, evaluation, and risk and ethics.

Evaluation hypotheses. Because DSR is governed by design-oriented research questions, hypotheses are limited to the quantitative evaluation of the artifact rather than used as the governing logic of the full study.

Table 3

Testable Hypotheses for the Evaluation Phase

Code	Testable hypothesis
H1	Within the structured evaluation context, the intelligent AAC prototype will produce a higher rate of independent functional communication acts per session than the predefined static AAC condition.
H2	Within the structured evaluation context, the intelligent AAC prototype will increase communicative diversity and reduce the level of prompting or assistance required relative to the static AAC condition.

Note. Author's own elaboration. The hypotheses operationalize the quantitative evaluation of the intelligent AAC prototype against the predefined static AAC condition.

General objective: By the end of the 36-month doctoral research period, design, develop, and rigorously evaluate a human-in-the-loop multimodal AAC prototype in Spanish that supports independent functional communication for the defined user group within one structured therapeutic context, while preserving user agency, privacy, and safety.

Specific objectives:

- Identify and formalize the functional, contextual, ethical, and accessibility requirements of the prototype through evidence review and focused participatory input.

- Design and implement a multimodal AAC prototype that integrates photographs/pictograms, text, speech output, contextual information, and human validation of AI-generated suggestions.
- Evaluate the prototype against a predefined static AAC condition using repeated within-participant observations and prespecified measures of independent communication, communicative diversity, and assistance required.
- Assess privacy, autonomy, bias, safety, usability, and potential communication-homogenization risks and refine the artifact in response to evidence.
- Derive transferable design principles for responsible, context-aware, human-in-the-loop personalization in Spanish-language AAC systems.

06. Methodology

Philosophical position and research approach. The study is anchored in Pragmatism and uses Design Science Research as the governing research approach. Knowledge is treated as valuable when it supports defensible understanding and improvement of action in a real problem context. DSR is appropriate because the doctoral contribution is expected to include both an evaluated technological artifact and transferable design knowledge.

DSR process. The study will follow a build–evaluate–iterate logic informed by the design-science literature. The protocol operationalizes this logic through six linked activities: (1) problem identification and motivation; (2) definition of solution objectives and requirements; (3) design and development of the artifact; (4) controlled demonstration in the target context; (5) repeated evaluation and refinement; and (6) communication of the artifact, evidence, limitations, and design principles.

Artifact. The artifact will be a Spanish-language multimodal AAC prototype that combines photographs or pictograms, text, speech output, contextual information, and a human-in-the-loop suggestion mechanism. The system may propose contextually relevant communication options, but the user must remain able to select, reject, modify, ignore, or stop the interaction. The system will not be designed to infer thoughts, diagnose the child, or communicate autonomously on the child’s behalf.

Participants and context. The pilot evaluation will target a minimum of four participants from one clearly defined subgroup of minimally speaking or nonspeaking

autistic children requiring very substantial support. Evaluation will occur in one structured therapeutic communication context. Final eligibility criteria, recruitment procedures, and access arrangements will be defined before ethics submission and data collection.

Participatory input. Caregivers, speech-language professionals, and autistic AAC users, including autistic adults where appropriate, will provide focused input on communication requirements, acceptability, risk, and contextual relevance. Their feedback will inform design and refinement without converting the study into a second governing methodology.

Evaluation design. The leading evaluation option is an alternating-treatments single-case design comparing the intelligent prototype with a predefined static AAC condition. Conditions will be presented in counterbalanced or randomized order where feasible, with repeated measurement across sessions. The protocol will prespecify the comparator, session procedures, prompting hierarchy, fidelity checklist, safety and stopping rules, and analysis plan.

Table 4

Outcome and Process Measures for Prototype Evaluation

Measure	Role in evaluation
Independent functional communication acts per session	Primary outcome; distinguishes independent/spontaneous acts from prompted acts.
Communicative diversity	Primary outcome; captures the range of communicative functions or concepts expressed.
Assistance / prompting level	Primary outcome; captures the amount of external support required to communicate.
Message-selection time	Secondary descriptive measure; faster interaction will not automatically be interpreted as better communication.
Rejected or modified AI suggestions	Process measure for user agency, relevance, and artifact refinement.

Note. Author's own elaboration. The measures distinguish primary communication outcomes, a secondary efficiency measure, and process indicators related to user agency and artifact refinement.

Data sources and analysis. Data will include prototype interaction logs, coded functional communication acts, communicative diversity, prompting level, rejected or modified AI suggestions, and focused caregiver or specialist feedback. Individual participant trajectories will be examined before aggregate summaries. Repeated within-participant comparisons will be emphasized, and a prespecified subset of sessions will be independently double-coded to estimate inter-observer agreement. Treatment fidelity will also be documented. Quantitative results will be interpreted as bounded pilot evidence rather than population-level clinical effectiveness.

Rigor and reproducibility. Before data collection, the protocol will define the artifact version, static comparator, inclusion criteria, primary outcomes, session structure, prompting hierarchy, coding rules, fidelity checks, stopping rules, analysis logic, and handling of protocol deviations. Major design changes during DSR iterations will be versioned and justified so that refinement remains transparent and traceable.

Feasibility boundary. The study intentionally avoids full clinical validation across home, school, and therapy settings. It will evaluate one prototype, one user subgroup, one context, and two to three primary outcomes within the doctoral timeline. Simulation may be used before human evaluation for stress testing or failure scenarios, but it will not substitute for evidence about real-user utility.

07. Ethical Considerations

Ethical approval will be obtained before any participant recruitment or human-data collection. Written caregiver consent and developmentally and communication-appropriate child assent procedures will be required. For minimally speaking or nonspeaking participants, assent will include observable willingness to engage and predefined withdrawal or distress signals; participation will stop whenever those signals appear.

The study will follow data-minimization principles, collecting only information required to address the research questions. Identifiers will be separated from research records, access will be restricted, and retention and deletion rules will be defined in advance. On-device

processing will be preferred whenever technically feasible, particularly for photographs, voice-related data, and interaction logs. No identifiable participant information will be stored in public repositories.

Human agency is a core ethical requirement of the artifact. AI suggestions will be treated as options rather than decisions. Users and caregivers must retain the ability to reject, modify, ignore, or stop interaction. The protocol will also monitor possible bias, culturally inappropriate vocabulary, over-personalization, unsafe suggestions, and homogenization of communication. Any event indicating distress, privacy risk, or unsafe system behavior will trigger predefined review or stopping procedures.

08. Expected Results

The expected results are stated as plausible outcomes rather than guaranteed findings. First, the project is expected to produce a functional Spanish-language human-in-the-loop multimodal AAC prototype whose architecture and interaction rules are explicitly documented.

Second, the evaluation is expected to generate repeated within-participant evidence about whether the artifact is associated with changes in independent functional communication, communicative diversity, and assistance required relative to a static AAC condition.

Third, the study is expected to identify design trade-offs involving contextual relevance, user agency, privacy, safety, and personalization. These findings should support a set of transferable design principles for responsible AAC personalization. Negative, null, or heterogeneous findings will also be considered scientifically valuable because they can reveal limits, failure modes, or conditions under which the artifact should not be used. The thesis will therefore avoid assuming that the prototype must outperform the comparator.

09. Timeline & Budget

Timeline. The work is planned across a 36-month doctoral research period, with explicit time for ethics approval, iteration, rework, and contingency rather than assuming that the first artifact version will succeed.

Table 5*Proposed 36-Month Research Timeline and Milestones*

Period	Main activities	Milestone
Months 1–6	Consolidated protocol; systematic literature review; requirements; ethics preparation; initial participatory input.	Approved protocol and ethics submission.
Months 7–12	Architecture, multimodal interaction design, data-flow design, prototype alpha, technical verification.	Prototype alpha and documented requirements.
Months 13–18	Usability/technical pilot; failure-mode testing; artifact refinement; finalize evaluation procedures.	Evaluation-ready prototype.
Months 19–24	Recruitment after approval; baseline/repeated sessions; begin comparative single-case evaluation.	Initial empirical dataset and fidelity evidence.
Months 25–30	Complete evaluation; double-coding subset; analysis; focused stakeholder interpretation; refine artifact.	Evaluation results and refined artifact.
Months 31–34	Synthesize design principles; prepare journal/conference manuscripts; integrate limitations and ethics findings.	Draft publications and contribution chapter.
Months 35–36	Integrate thesis, final quality review, repository/documentation cleanup, defense preparation.	Final dissertation and defense package.

Note. Author’s own elaboration. The timeline organizes the doctoral research into sequential phases covering protocol development, artifact design, pilot testing, empirical evaluation, analysis, dissemination, and dissertation completion.

Preliminary budget. The estimate below is intentionally conservative and will be refined once institutional access, available equipment, and software arrangements are confirmed. It excludes university tuition and the researcher's own labor.

Table 6

Preliminary Budget for the Design, Development, and Evaluation of the AAC Prototype

Item	Estimated amount (S/)	Rationale
Prototype devices and accessories	3,000	Tablet/device accessories, microphones, protective cases, testing peripherals, replacements.
Cloud / development services	1,800	Limited cloud inference, storage, development or API costs when on-device processing is insufficient.
AAC / localization resources	600	Peruvian Spanish voice, icon/pictogram adaptation, printing or accessibility materials where open resources are insufficient.
Field logistics and local transport	1,800	Researcher travel to the structured evaluation site and session logistics.
Research materials / secure data management	900	Encrypted storage, backups, coding materials, documentation, secure media.
Participant-session support / incidental materials	900	Non-coercive session materials and minor accessibility/comfort items,

		subject to ethics approval.
Contingency (~15%)	1,350	Unexpected replacement, iteration, re-testing, or logistics costs.
Preliminary total	10,350	Subject to refinement after access and ethics arrangements.

Note. Author's own elaboration. The budget provides a preliminary estimate of resources required for prototype development, localization, secure data management, field logistics, participant-session support, and contingency costs. Amounts are subject to refinement after confirming institutional access, available equipment, ethics requirements, and technical infrastructure.

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